

Two Distributions, Opposite Verdicts: Feature Provenance in Cross-Dataset Federated Intrusion Detection

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Abstract—Cross-dataset intrusion-detection results are usually attributed to distribution shift between corpora. We show that for a fixed source model and a fixed set of target flows, the published verdict can instead be decided by which public CSV distribution of the target dataset supplied the features. We train logistic-regression FedAvg on InSDN in a five-client, five-seed setting and evaluate it zero-shot on CICIDS2017. The primary experiment is a row-matched counterfactual: the same 207,463 target flows in the same order, with identical labels, identical attack prevalence, and seven bit-identical features, in which only `Protocol` and `Source Port` are replaced by the surrogates that the `MachineLearningCVE` release forces. ROC-AUC falls from 0.7357 to 0.3545 and MCC from +0.2511 to -0.1915, a below-random cross-domain ranking that could otherwise be misread as model-transfer failure. The actual `MachineLearningCVE` release reproduces this (0.3441, MCC -0.2698), establishing practical relevance, and a centralized model trained on the same records behaves the same way (0.7490 $\rightarrow$ 0.3544), so the effect is not an artifact of FedAvg aggregation. Masking localizes the mechanism to protocol, whose learned contribution changes sign (-0.1438 versus +0.2647 ROC-AUC) while its marginal Kolmogorov-Smirnov distance is the smaller of the two; on identical rows, dropping the unavailable field is markedly safer than reconstructing it (0.5157 versus 0.3545). A label-free preflight screen flags the substituted port on every seed, but none of the tested marginal distance rules separates the reconstructed protocol, so feature provenance has to be reported per field. All mirrors are regenerated from public files by the released script and match published SHA-256 checksums.

Index Terms—Federated learning, intrusion detection, cross-dataset evaluation, feature provenance, reproducibility.

I. INTRODUCTION

Federated learning (FL) lets participating sites improve a shared intrusion detector without centralizing raw records [1], [2]. Cross-dataset tests are the standard check on whether such a model generalizes beyond its training corpus, and they routinely report severe degradation [3]–[6], normally explained by differences in traffic, attack mix, or labeling.

This paper reports a different and largely unexamined cause. Widely used corpora ship in more than one flow-feature distribution, and those distributions do not expose the same columns. CICIDS2017 has a `TrafficLabelling` release retaining `Protocol` and `Source Port`, and a `MachineLearningCVE` release omitting both, so anyone

harmonizing feature names against the second must reconstruct protocol and substitute a placeholder port. We show that this choice, not the target traffic, can decide whether transfer looks usable or below random.

We ask three questions: (RQ1) does FedAvg improve over matched centralized and local controls; (RQ2) how does an explicit non-IID partition affect ranking and fixed-threshold measures; and (RQ3) does the verdict depend on which public distribution supplied the target features, and which field drives that dependence? Claims are restricted to outputs of the accompanying script.

Our contributions are: (i) matched centralized, local-only, IID, and non-IID controls with paired seed-level reporting; (ii) a row-matched counterfactual that isolates the mechanism, holding the target rows, their order, their labels and all seven remaining features bit-identical while only the two provenance-sensitive fields change, paired with the real `MachineLearningCVE` release as an ecological check and with a centralized control showing the effect is not specific to federated averaging; and (iii) a masking diagnostic that localizes the reversal to protocol, a quantified comparison showing that dropping an unavailable field is safer than reconstructing it, and a label-free preflight screen that catches placeholder substitution but not this reconstruction, which is why per-field provenance must be reported. All mirrors are rebuilt from public files by the released script.

II. RELATED WORK AND RESEARCH GAP

A. Federated intrusion detection and client heterogeneity

FedAvg established sample-weighted aggregation of local updates [1], and security surveys distinguish distributed optimization from formal privacy or attack robustness [2]. Because non-IID clients cause objective drift, several methods address it directly: FedProx adds a proximal term [7], SCAFFOLD corrects drift with control variates [8], and FedNova normalizes unequal local work [9]. FL-IDS work spans linear scanning detectors [10], 5G experiments with centralized and local controls [11], and deep or distillation-based defenses [12], [13]. Dataset-centric FL evaluation reports in-domain scores approaching centralized ones while cross-dataset transfer degrades

sharply, and recommends documenting feature harmonization rather than comparing headline accuracy [14]. We keep the optimizer at plain FedAvg and the model at logistic regression, so the aggregation variants above are the natural next step rather than a claim here.

B. Cross-dataset NIDS and feature semantics

Dataset surveys show that collection environment, labeling, traffic generation, and flow extraction constrain valid comparisons [15]. Inter-dataset studies report large degradation even between related CIC corpora [3]; XeNIDS formalizes cross-evaluation use cases and warns against assuming interchangeable flow fields [4]. Later work finds asymmetric transfer across four datasets [5] and near-perfect within-domain yet poor cross-dataset performance [6]. Standardized NetFlow feature sets improve generalization but feature influence stays dataset-dependent [17], and domain adaptation exists precisely because feature spaces rarely align [18]. Closest to our setting, Demirbağ Paray and Aydos harmonize CICIDS2017 and InSDN/OVS into a common schema for multi-domain federated IDS, justifying it on the ground that both corpora were produced by the same extractor, so that fields such as flow duration and packet length “carry identical semantic and technical definitions” across environments [16]. Our results qualify that premise: shared extractor lineage and matching column names do not by themselves guarantee semantic interchangeability, because different public releases of the *same* corpus expose or reconstruct fields differently enough to reverse the verdict.

C. Dataset defects, tooling, and release variants

CICIDS2017 has documented defects rather than merely different statistics. Engelen et al. report flow-construction and labeling faults [19]; Lanvin et al. find packet misordering, duplication, and an unlabeled port scan, and measure the resulting change in detection performance [20]; Rosay et al. audit the released flows and report duplicated features, miscalculated features, and *wrong protocol detection*, noting that the published CSVs cannot be regenerated from the PCAPs with any available extractor version [21].

This literature establishes that a corpus can be internally faulty and that two corpora are rarely comparable. The gap we address is narrower and, to our knowledge, unquantified: a *single* corpus is distributed in several feature releases exposing different columns, and the release chosen for the target side can invert the sign of a cross-dataset conclusion. Comparing two releases directly cannot settle this, because their row sets and attack prevalence also differ. We therefore construct a counterfactual on one fixed row set so that feature provenance is the only variable, then report the mechanism and ask what an automated check could have caught.

III. METHOD

FedAvg aggregates locally updated parameter vectors as $\mathbf{w}_{t+1} = \sum_{k=1}^K (n_k / \sum_j n_j) \mathbf{w}_{t+1}^k$ with $K = 5$ clients and local sample counts n_k [1]. Each client runs one local pass of binary logistic regression per round, producing $p(y = 1 |$

$\mathbf{x}) = \sigma(\mathbf{w}^\top \mathbf{x} + b)$. The model is a transparent tabular baseline, not a deep sequence detector, chosen so that the cross-view comparison cannot be attributed to architecture.

Server and clients begin from the same neutral zero coefficient/intercept state, so no client record is used for initialization, and a “local epoch” is one complete `partial_fit` pass rather than a mini-batch count. Centralized LR uses the same optimizer and 20 passes; local-only LR keeps independent client states and averages their test metrics. The source-training mean and standard deviation define each z-score and are retained unchanged for the InSDN test partition and all three target views; non-finite rates and missing values become zero before that fit.

A. Feature-shift diagnostic

For source feature j with empirical CDF F_{sj} and target CDF F_{tj} , we record the two-sample Kolmogorov–Smirnov distance $D_j = \sup_x |F_{sj}(x) - F_{tj}(x)|$. This marginal statistic cannot reveal whether a learned association changes sign. We therefore keep the trained IID global model fixed, replace one standardized feature by zero (the source-training mean) in both test domains, and compute $\Delta A_j = A_t^{\text{mask}(j)} - A_t^{\text{full}}$ for target ROC-AUC. A positive ΔA_j identifies a feature whose learned contribution harms target ranking; a negative one identifies a feature carrying transferable signal. For a linear score the masked logit is exactly $\mathbf{w}^\top \mathbf{x} - w_j x_j$, so the diagnostic is cheap and exact here, and it applies unchanged to nonlinear scores. It does, however, consume target labels, so it audits a finished experiment and cannot protect one in progress.

B. Label-free provenance screen

We therefore ask what can be established before any target label is available, using only the source-training column x_j and the unlabelled target column $\tilde{x}_j$. The screen raises a flag when either

$$\frac{\text{sd}(\tilde{x}_j)}{\text{sd}(x_j)} < \tau_v \quad \text{or} \quad \Pr[\tilde{x}_j \notin [\min x_j, \max x_j]] > \tau_r, \quad (1)$$

with $\tau_v = \tau_r = 0.01$. The first condition detects a column that has lost essentially all variation, the second a column whose values were never seen in training; a placeholder written for an absent field triggers both. The thresholds are deliberately one-sided, since a *large* dispersion ratio is ordinary shift rather than missing data. For integer-coded fields we also record two candidate rules: the total-variation distance between category proportions, and whether any category holding over 5% of source mass falls below 1% of target mass. The label-aware ΔA_j is the reference signal these label-free rules are compared against, not causal ground truth: with correlated features, masking reports how the model relies on a column, not how much it was corrupted.

IV. EXPERIMENTAL PROTOCOL

A. Source data and the two target mirrors

We use InSDN [22] for training and within-domain testing. The prepared InSDN file contains 343,889 records (68,424

TABLE I
THE THREE TARGET VIEWS OF THE SAME CAPTURE FILES. CF IS ROW-MATCHED TO TL BY CONSTRUCTION; ML IS THE REAL RELEASE AND DIFFERS IN ROW SET AS WELL.

	TL (original)	CF (counterfactual)	ML (MLCVE)
Protocol field	original	flag surrogate	flag surrogate
Protocol values	{0, 6, 17}	{0, 6}	{0, 6}
Source port	original	sentinel -1	sentinel -1
Cleaned flows	1,037,315	1,037,315	880,176
Fixed test split	207,463	207,463	176,036
Sampled benign/attack	21,705 / 8,295	21,705 / 8,295	22,577 / 7,423
Attack prevalence	0.2765	0.2765	0.2474
Row-matched to TL	—	yes	no

normal; 275,465 anomaly) and carries true IANA protocol numbers {0, 6, 17}. Each seed draws a stratified 30,000-record sample, split into 24,000 train and 6,000 test records, and draws 30,000 records from each target view.

All three target views describe the *same* CICIDS2017 capture files [23]: Monday benign, Friday DDoS, Friday PortScan. The TL view comes from TrafficLabelling and keeps the original Protocol and Source Port fields. In MachineLearningCVE neither column exists, so protocol must be reconstructed and the port replaced. We evaluate one plausible but lossy strategy: protocol becomes TCP (6) when any TCP flag is positive and 0 otherwise, and every source port becomes -1. This is a choice, not a standard, and Section V-B contrasts it with dropping the field altogether.

Comparing TL against ML alone would be confounded, because the two releases survive cleaning and de-duplication differently and end up with different row sets and attack prevalence (Table I). The CF view removes that confound: it is derived from the TL test split itself, the same 207,463 rows in the same order with the same labels and same seven remaining features, and only `ip_proto` and `tp_src` overwritten by the ML surrogates. Deriving it from the finished row set rather than re-running the cleaner is essential, since the missing-value filter and the de-duplication both key on the two columns being changed and would otherwise select different rows. The preparation script asserts that the unaffected columns and labels are bit-identical and that exactly two columns differ; the evaluation script repeats that assertion after seeded subsampling, and either failure aborts the run. The surrogate assigns a different protocol to 30.32% of rows, since UDP (17) has no representation in the flag-derived rule.

The complete input is nine columns, identically named in the prepared InSDN CSV and in every target view and referenced by name rather than by position: `ip_proto`, `tp_src`, `tp_dst`, `packet_count`, `byte_count`, `duration_sec` (seconds), `packet_count_per_sec`, `byte_count_per_sec`, and `packet_size_avg` (bytes). All nine are z-scored; the two rate columns have non-finite values mapped to zero beforehand; ports span 0–65,535 except for the sentinel -1; protocol is integer-coded and is one-hot encoded only in the corresponding ablation, which also removes both ports. The prepared CSVs additionally carry

a `flow_duration` column identical to `duration_sec` in every row, an instance of the feature duplication reported for these releases [21]; it is excluded so that no coefficient is split across duplicated inputs. Labels are reduced to binary: InSDN `normal`→0 and `anomaly`→1, giving mean sampled counts of 5,969 benign and 24,031 attack, while in every target view `normal`→0 and `ddos/portscan`→1. Sampling occurs after label conversion and numeric cleanup and is stratified to retain prevalence. No SMOTE output, source string, run identifier, or label field is supplied to the classifier.

StandardScaler is fit once on the InSDN training features and applied unchanged to the InSDN test partition and all three target views; no target-domain fit, fine-tuning, or threshold selection is performed.

B. Partitions and measures

For IID FL, shuffled training records are divided equally among five clients. For non-IID FL, each class is allocated using a Dirichlet distribution with $\alpha = 0.5$. Both configurations run 20 rounds fixed *a priori*; no early stopping, model selection, or tuning uses the test set. We report precision, recall, F1, ROC-AUC, PR-AUC, balanced accuracy, MCC, and confusion matrices. The random PR baseline equals the attack prevalence: 0.801 in InSDN and the mirror values in Table I.

Predicted labels use the fixed 0.5 threshold; ROC-AUC and PR-AUC use continuous scores. Balanced accuracy stops the majority attack class from dominating, and MCC uses all four confusion cells, its sign making a below-random operating point visible. The saved CSV records every metric per seed, setting and target.

IID partitioning gives every client 4,800 records and retains class proportions, whereas the Dirichlet split produces severe skew: at seed 11, client sizes range from 1,669 to 11,950 records and the benign share from 0.5% to 38%. Every partition is recorded in the artifact. Centralized LR uses the identical 24,000 training records; local-only LR trains five independent client models and reports their unweighted metric mean.

The classifier is a scikit-learn `SGDClassifier` with `log_loss`, constant-rate SGD at $\eta = 0.01$ and L2 $\alpha = 10^{-4}$, no class weighting, and a fixed 0.5 threshold; there are $K = 5$ clients, 20 rounds, one full local pass per round, seeds 11, 23, 42, 77 and 101, and infinities are mapped to NaN and then to zero before scaling. The pinned environment is Python 3.11 with NumPy 2.2.6, pandas 2.3.1 and scikit-learn 1.7.2, and `summary.json` records every one of these values alongside the results.

V. RESULTS

A. Source-domain controls

Table II reports mean±std over five seeded repeated holdouts. All four configurations reach F1 near 0.956–0.959, with FedAvg IID between the centralized and local-only controls and a gap to centralized smaller than the seed-to-seed deviations. Local-only has the highest mean F1, which is not evidence of superiority since it is an unweighted mean over five models on one test set. Non-IID allocation preserves ROC-AUC while

TABLE II
SOURCE-DOMAIN (INSDN) FIVE-SEED RESULTS; MEAN $\pm$ STD.

Setting	F1	ROC-AUC	Bal. acc.	MCC
Central	.9567 $\pm$.0033	.9320 $\pm$.0038	.8285 $\pm$.0149	.7624 $\pm$.0199
Local-only	.9588 $\pm$.0027	.9288 $\pm$.0030	.8367 $\pm$.0120	.7751 $\pm$.0162
FedAvg IID	.9580 $\pm$.0023	.9324 $\pm$.0039	.8355 $\pm$.0091	.7709 $\pm$.0140
FedAvg non-IID	.9556 $\pm$.0038	.9336 $\pm$.0064	.8136 $\pm$.0166	.7562 $\pm$.0228

reducing balanced accuracy and MCC, an operating-point effect under client class skew rather than uniformly worse ranking. Round 20 was fixed before evaluation, so no round-wise quantity controls stopping or tuning; the saved history CSV shows both partitions flat well before it. For seed 11, the IID FedAvg confusion matrix contains TN=833, FP=361, FN=33, and TP=4,773, so the class imbalance explains why F1 and PR-AUC alone are insufficient and why Table II includes MCC. Per-seed values are in `five_seed_metrics.csv`: IID FedAvg F1 ranges from 0.9557 to 0.9606 and the non-IID range is wider (0.9510–0.9611), as expected when both sampled records and client allocations change.

B. Two fields reverse the verdict on identical rows

The same five IID FedAvg models are now scored on all three target views without any refitting (Table III). On TL, transfer is clearly degraded relative to the source domain (0.9324 $\rightarrow$ 0.7357 ROC-AUC) but remains informative, with positive MCC and PR-AUC 0.4520 against a 0.2765 prevalence baseline. Overwriting only `ip_proto` and `tp_src` on those very rows sends every measure below chance: ROC-AUC 0.3545 and MCC -0.1915 . Because CF shares its rows, order, labels, prevalence and seven features with TL, this difference cannot be attributed to attack mix, labeling, collection environment, or sampling; the only variable is which values those two columns hold. The seed-to-seed standard deviations are an order of magnitude smaller than the gap.

The real ML release lands close to the counterfactual (0.3441 against 0.3545), the ecological half of the argument: the mechanism isolated on matched rows dominates in an actual public release, even though that release also differs in row set and prevalence. Reported alone, that row is a below-random result which would ordinarily be read as model-transfer failure. The centralized model shows the same pattern (0.7490 $\rightarrow$ 0.3544 $\rightarrow$ 0.3447), so this is a property of feature semantics rather than of federated averaging, and the hazard applies to any classifier trained on one release and evaluated on another.

C. Localizing the reversal

Table IV perturbs the fixed model one feature at a time. Protocol is the only field whose masking effect changes sign, and it does so on matched rows: on TL it carries the largest transferable contribution, with masking costing 0.1438 ROC-AUC, whereas on CF masking the surrogate *gains* 0.2647. The real release agrees (+0.2103). Marginal shift magnitude does not identify this. The KS distance for protocol is 0.3532 on

TABLE III
ZERO-SHOT CROSS-DATASET RESULTS, FIVE SEEDS, NO REFITTING. CF DIFFERS FROM TL IN EXACTLY TWO COLUMNS ON IDENTICAL ROWS.

Target view	F1	ROC-AUC	PR-AUC	Bal. acc.	MCC
<i>FedAvg IID global model</i>					
TL (original)	.5012 $\pm$.0037	.7357 $\pm$.0070	.4520 $\pm$.0154	.6403 $\pm$.0037	.2511 $\pm$.0065
CF (row-matched)	.3558 $\pm$.0026	.3545 $\pm$.0031	.2102 $\pm$.0009	.4191 $\pm$.0037	-.1915 $\pm$.0097
ML (real release)	.2914 $\pm$.0018	.3441 $\pm$.0040	.1848 $\pm$.0010	.3766 $\pm$.0023	-.2698 $\pm$.0067
<i>Centralized model, same records</i>					
TL (original)	.4950 $\pm$.0041	.7490 $\pm$.0165	.4656 $\pm$.0225	.6336 $\pm$.0047	.2392 $\pm$.0083
CF (row-matched)	.3545 $\pm$.0020	.3544 $\pm$.0033	.2100 $\pm$.0010	.4163 $\pm$.0045	-.2004 $\pm$.0147
ML (real release)	.2903 $\pm$.0041	.3447 $\pm$.0048	.1848 $\pm$.0012	.3738 $\pm$.0074	-.2782 $\pm$.0198

TABLE IV
MODEL-PRESERVING FEATURE AUDIT; FIVE-SEED MEANS. Δ AUC IS MASKED MINUS FULL ROC-AUC ON THAT VIEW. SOURCE F1 IS SHARED BECAUSE THE SOURCE MODEL IS IDENTICAL.

Masked feature	Src. F1	TL		CF		ML	
		KS	Δ AUC	KS	Δ AUC	KS	Δ AUC
Protocol	.9070	.3532	-.1438	.0992	+.2647	.0992	+.2103
Destination port	.9611	.3533	-.0605	.3533	-.0542	.3535	-.0583
Mean packet size	.9500	.5642	+.0547	.5642	+.0230	.5571	+.0639
Source port	.9563	.4287	+.0165	1.0000	.0000	1.0000	.0000

TL and only 0.0992 on CF, so the harmful encoding is the one that looks better aligned, even though the two are computed on the very same rows. Source port shows the converse failure: its CF distance is maximal because every value is the sentinel, yet masking it changes nothing, since a constant column contributes a constant to the logit and cannot affect ranking.

Retraining confirms the mechanism rather than inferring it from perturbation alone, and yields the comparison that matters when a column is unavailable. Table V removes fields before training. Dropping protocol costs source F1 (0.9580 $\rightarrow$ 0.9118) and collapses all three views to near chance (0.4953, 0.5157, 0.4668): once protocol is gone the provenances agree, because the field they disagree about is unused. Crucially, on the matched rows of CF, *dropping* the unavailable field reaches 0.5157 while *reconstructing* it reaches only 0.3545, so incorrect reconstruction is substantially worse than admitting the field is missing. That inverts the intuition behind name-based harmonization. One-hot encoding protocol is the best TL configuration (0.7472) yet barely moves CF (0.3630), so a representation change cannot repair reconstructed values. Removing both ports lowers TL ROC-AUC to 0.7062 with source F1 unchanged, indicating genuine but secondary port signal the ML release cannot express.

D. What a label-free preflight check can and cannot catch

Table VI applies the check of Section III-B, which sees no target label. Among the 27 evaluated column–view pairs it flags two on all five seeds and no others: the source port of CF and of ML, which retain none of their variation (ratio 0.0000) and lie entirely outside the source-training range. The separation is not marginal, since no unflagged column falls below ratio 0.352 or exceeds out-of-range 0.0028. The one-sided thresholds are justified in the same table, where the byte-rate column is 31 to 48 times more dispersed yet has Δ AUC 0.0000. We claim no

TABLE V
RETRAINING ABLATIONS, FIVE SEEDS. SOURCE COLUMNS ARE INSDN;
VIEW COLUMNS ARE ZERO-SHOT ROC-AUC.

Features	Src. F1	Src. AUC	TL	CF	ML
All nine	.9580±.0023	.9324±.0039	.7357	.3545	.3441
Without ports	.9583±.0030	.9267±.0065	.7062	.3005	.2857
Without protocol	.9118±.0042	.8915±.0067	.4953	.5157	.4668
One-hot protocol	.9582±.0023	.9311±.0036	.7472	.3630	.3506

TABLE VI
LABEL-FREE CHECK, FIVE-SEED MEANS. RATIO IS TARGET OVER
SOURCE-TRAINING STANDARD DEVIATION; OOR IS THE FRACTION OF
TARGET ROWS OUTSIDE THE SOURCE-TRAINING RANGE. A FLAG REQUIRES
RATIO < 0.01 OR OOR > 0.01; FLAGGED CELLS ARE BOLD.

Column	TL		CF		ML	
	Ratio	OOR	Ratio	OOR	Ratio	OOR
Protocol	1.0401	.0000	0.5672	.0000	0.5606	.0000
Source port	0.9481	.0028	0.0000	1.0000	0.0000	1.0000
Destination port	1.1984	.0005	1.1984	.0005	1.2650	.0006
Byte rate	30.6377	.0023	30.6377	.0023	47.6245	.0028

more than a cheap preflight sanity check: detecting a constant placeholder is easy, and what makes it worth reporting is the field it does *not* flag.

That field is the one that reverses the verdict. The reconstructed protocol keeps a plausible dispersion ratio (0.5672 on CF) and never leaves the source range, so both conditions stay silent. This is not because the corruption leaves no statistical trace; it does, and the table shows it. The trace simply fails to distinguish semantic corruption from ordinary shift under the marginal rules we tested. Total-variation distance repeats the KS failure, scoring the harmful encoding as the *closer* one (0.1489 against 0.3532) despite identical rows, and the mass-collapse rule fires on all three views, including uncorrupted TL, where protocol 0 falls from 35% of source mass to 0.04%. A rule firing on the intact view as readily as on the damaged ones cannot discriminate, so we report that negative result rather than tune thresholds until the cases separate.

The consequence is a division of labour. Placeholder substitution is statistically obvious and a preflight check should reject it, whereas a semantically reconstructed field may retain plausible marginal statistics and evade such checks, so it can only be known from provenance metadata recorded at preparation time. That is why our script records, per input file, whether each such field was read or reconstructed.

VI. DISCUSSION

A. What the controls establish

The controls share the feature set, records, optimizer family, class weight and threshold, so their close source-domain scores establish only that this linear baseline is adequate for the sampled InSDN task, not that federation improves accuracy. The non-IID experiment makes a narrower point: under the skew of Section IV-B, mean F1 falls only slightly while balanced accuracy and MCC fall visibly, so reporting all three stops high attack prevalence from hiding an unfavorable tradeoff.

Severe non-IID settings, drift-corrected aggregation [7]–[9], client dropout, Byzantine clients, privacy accounting and communication cost are outside this experiment.

B. Why provenance, not traffic, is the operative variable

The TL–CF contrast is what licenses a causal reading. Those views share rows, order, labels, prevalence and seven of nine features, and are scored by the same models under the same source-fitted scaler and threshold, so the difference in Table III cannot be attributed to attack mix, labeling, collection environment, or sampling; only two columns differ. The TL–ML contrast alone would not support that reading, since those releases also differ in row set and prevalence; its role is to show the isolated mechanism dominating in a release people actually download.

The effect is a reversal rather than a loss because the model applies a coefficient to a field whose target values now encode something else. Protocol becomes a TCP-flag indicator omitting UDP entirely, disagreeing with the original on 30.32% of matched rows, so its contribution is not merely uninformative but anti-correlated, which drives MCC below zero. Rosay et al. independently report wrong protocol detection in these releases [21]; we quantify what such a field does to a verdict. The ablations add the corollary: when a meaningful field is unavailable, dropping it degrades transfer to near-chance, but reconstructing it incorrectly is considerably worse.

Section V-D states the limit of automation. A preflight check catches placeholder substitution and should be run, but it did not catch the reconstruction in this case study, and none of the marginal distance rules we tested separates the two protocol encodings even on identical rows. A cross-dataset NIDS result must therefore state which release of each corpus supplied each feature and which fields were reconstructed rather than read. Absent that record, a below-random result cannot be distinguished from an artifact of harmonization, and neither can a favorable one.

The external evaluation fixes every source-fitted quantity, including scaling and the 0.5 threshold, which is stricter than re-fitting either on target labels. The residual TL degradation still admits several causes, including attack mix, label reduction, and the limited expressiveness of nine aggregate features; this design does not attribute it to any one of them.

VII. THREATS TO VALIDITY AND REPRODUCIBILITY

The experiment evaluates tabular flows, not packet traces, and is not an SDN deployment. Five repeated holdouts share a finite source corpus and are not independent sites. The contrast covers one source corpus, one target corpus, two model families, and three capture files; whether the reversal generalizes to other release pairs is open. The reconstruction we evaluate is one plausible rule, the obvious one when only flag counts survive; another rule would give other magnitudes. The counterfactual establishes what these two fields do to this model on these rows, not that provenance is the largest source of cross-dataset error in general. The preflight thresholds were fixed at 0.01 rather than tuned, and the check is one-directional:

a flag is strong evidence of substitution, but silence is not evidence of sound provenance. Masking is a label-aware model-reliance diagnostic, not a causal attribution, since correlated features share credit. Binary reduction limits external validity. The artifact preserves construction code, per-file provenance, checksums, seeds, partitions, metrics and histories.

VIII. ARTIFACT AVAILABILITY

The preparation script converts named public source files, records which branch supplied each reconstructed field, publishes SHA-256 checksums for the InSDN file and each target view, and aborts if the counterfactual is not row-matched. The evaluation script regenerates the five-seed outputs, the three-view audit and the preflight check under the pinned Python 3.11 environment, and a companion script re-reads this paper and fails if any reported value disagrees with the saved outputs. The artifact is at <https://github.com/thtsec/rivf2026-federated-edge>; third-party datasets are not redistributed.

IX. CONCLUSION

Matched controls show no FedAvg accuracy advantage on sampled InSDN, while non-IID skew primarily harms fixed-threshold measures. On one fixed set of 207,463 target flows, with identical labels, prevalence and seven identical features, overwriting only protocol and source port with the surrogates a public release forces moves ROC-AUC from 0.7357 to 0.3545 and MCC from +0.2511 to -0.1915. The real MachineLearningCVE release reproduces the effect (0.3441) and a centralized model shows the same pattern, so this is a property of feature semantics rather than of federated averaging. Masking localizes it to protocol, whose smaller marginal shift is the more harmful encoding, and the ablations show that dropping an unavailable field is safer than reconstructing it wrongly (0.5157 versus 0.3545). A label-free preflight check catches the substituted port but not the reconstructed protocol, and none of the tested marginal distance rules separates the two encodings. Cross-dataset NIDS evaluations should therefore report feature provenance per field. Future work should test other corpora, release pairs, reconstruction rules, and model families.

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